Suad Arzimanli

Geospatial Software & GIS Developer
Łódź, Poland · Open to EU / EEA opportunities (onsite, hybrid, or remote)
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Profile

Geospatial software developer focused on public-infrastructure and municipal GIS: C#, .NET, ArcGIS (Pro, Online, Experience Builder), Cityworks AMS, and 311/Siebel integrations. Delivers end-to-end systems from database and backend design through secure APIs, IIS deployment, and legacy modernisation. Based in Poland; open to EU/EEA GIS, GeoAI, and geospatial engineering roles. Working languages: English (professional); German (intermediate).

Professional Experience

Full-Stack .NET and GIS Developer — *2023 – Present*

DZ CIS — USA (Remote)

Onboarding, training & foundation

- Started in a trainee/intern capacity with hands-on training in Cityworks AMS, enterprise GIS workflows, ArcGIS Pro, and geodatabase structures.
- Supported senior developers on backend (.NET, SQL) and frontend (map-based web interfaces) for asset-management and service-request applications.
- Built practical experience with geospatial datasets, 311 service-request lifecycles, and municipal infrastructure systems.

Development & core engineering

- Developed full-stack enterprise geospatial asset-management applications (Google Maps API, .NET MVC, ArcGIS services) for Tree Management, 311/Pothole, and DOT/Highway operations.
- Designed and built an end-to-end municipal asset-management web application (.NET 8 / ASP.NET Core): ArcGIS asset inventory on Google Maps, Siebel/311 service requests via Azure APIM, SQL-backed inspections and work orders, and Entra ID role-based access.
- Implemented full-cycle field workflows: service-request–to-asset linking, map-based bulk work creation, attachments, edit locks, and PDF exports.
- Engineered interactive map tools: custom markers, clustering, radius filtering, overlays, geofencing, and location search.
- Built ArcGIS Experience Builder solutions and custom widgets; used ArcGIS Pro / Online in production workflows.
- Automated Cityworks AMS workflows (service requests, inspections, fieldwork) via API-integrated solutions.
- Integrated Cityworks, Siebel 311, ArcGIS Feature Services, and internal databases over REST.
- Designed and maintained enterprise geodatabases (integrity, schema consistency, operational readiness).
- Performed geospatial analysis, spatial queries, cleaning, validation, and QA/QC with SQL and Python.
- Developed modular C#/.NET backend logic (async operations, structured controllers) for performance and maintainability.
- Wrote technical documentation, workflow diagrams, and user guides for delivery and maintenance.

System ownership, security & modernisation

- Modernised .NET applications (framework migrations to newer .NET releases, project versioning).
- Aligned JavaScript map UI behaviour with ASP.NET Core MVC/Razor APIs and service-layer contracts.
- Deployed and maintained applications on IIS (configuration, environments, production stability).
- Maintained prod vs test environments (Siebel instance selects matching SQL and APIs); applied ArcGIS FeatureServer updates in production with edit locks / applyEdits.
- Implemented SSO, Azure AD / Microsoft Entra ID (OIDC, app roles), certificates (.pfx / .cert), and secure API communication.
- Maintained secure APIs and integrations using API management practices.

Technical leadership

- Technical lead (Team Lead): task assignment, coordination of GIS/.NET work, mentoring teammates.
- Led a team building ArcGIS Experience Builder applications for operational systems.
- Onboarding, training, and Q&A for teammates on map products, workflows, and spatial analysis (while continuing delivery).
- Designed and implemented an internal team management system for task tracking.

Collaboration & delivery model

- Agile/Scrum in a remote, cross-functional team (sprint planning, stand-ups, iterative delivery); Time & Materials engagement.

Technical Skills

Software: C#, .NET (MVC, Windows Forms, ASP.NET Core, Web API), EF Core, async processing, application lifecycle, legacy modernisation, SQL Server (schema design, backups, archives), IIS

Web / mapping: JavaScript, HTML/CSS, Google Maps API (markers, clustering, overlays, geofencing, search), Leaflet, Chart.js

GIS: ArcGIS Pro, ArcMap, ArcGIS Online, ArcGIS Experience Builder (custom widgets), Feature Services, enterprise geodatabases, projections

Operations GIS: Cityworks AMS (workflow automation, inspections, service requests), Siebel 311, Tree Management and DOT/Highway asset workflows

Data / QA: SQL (spatial and non-spatial), Python / ArcPy (processing, cleaning, QA/QC, automation), production GIS data reliability

Security / integration: REST APIs, system-to-system integration, SSO, Entra ID (OIDC, app roles), certificates, secure APIs, API management

Tools / EO: Visual Studio, Postman, Microsoft Office, Agile/Scrum, technical documentation; STAC, GeoParquet, Sentinel-2, Landsat, NDVI/LST, geospatial ETL/lineage (research)

Selected Projects & Research

GeoLineage *2025 – Present*

Independent open-source research

Provenance-first GeoETL: STAC ingest → GeoParquet; per-run lineage JSON, run-to-run input diff, minimum recompute. Validated on Element84 Earth Search and Copernicus Data Space with validation suite and reproducibility scripts. GitHub

Urban Protest-Count Forecasting *2025 – Present*

Collaborative research (Virginia Tech)

First author with S. Simsek and C.-T. Lu: integrating urban data for one-month-ahead forecasting of reported protest counts. *Manuscript prepared for submission to IEEE Big Data* (not yet submitted or accepted).

Urban Heat Island Analysis, Łódź *2025*

Independent EO study

Sentinel-2 NDVI and Landsat 8 LST on a 500 m grid; regression and spatial analysis; interactive Leaflet map; documented workflow.

GEOINT Crisis Mapping *2024 – 2025*

BSc thesis — University of Łódź

Web GIS for conflict-event analysis (Poland/ACLED); forecast panel (3-month moving average); temporal holdout evaluation of forecasting baselines (moving average, naive, Random Forest). ASP.NET Core, Google Maps API, Chart.js.

Faculty collaboration — dr Łukasz Lechowski *2025 – Present*

Institute of Urban Geography, Tourism and Geoinformation, University of Łódź

Collaboration recently started; no joint paper yet (distinct from the IEEE manuscript).

Education

MSc Mathematics and Computer Science *2025 – Present*

University of Łódź, Poland. Programme taught in English.

BSc Mathematics and Computer Science *Oct 2021 – July 2025*

University of Łódź, Poland. Thesis: "Application using Geospatial Intelligence (GEOINT) to map and analyze geographic data" — web GIS for conflict-event analysis and forecasting (ACLED).

Languages

- **English** — Professional working proficiency (C1/C2)
- **Azerbaijani** — Native
- **Turkish** — Native
- **German** — Intermediate (A2–B1); improving for DACH professional use